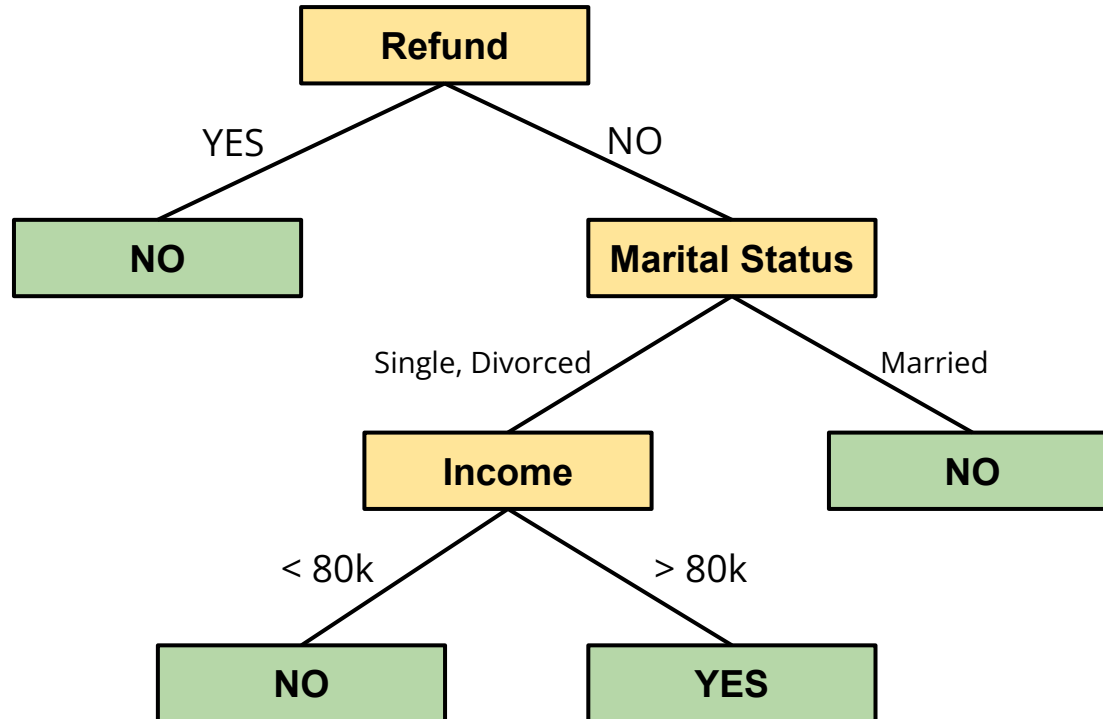

Decision Trees

— Boston University CS 506 - Lance Galletti —

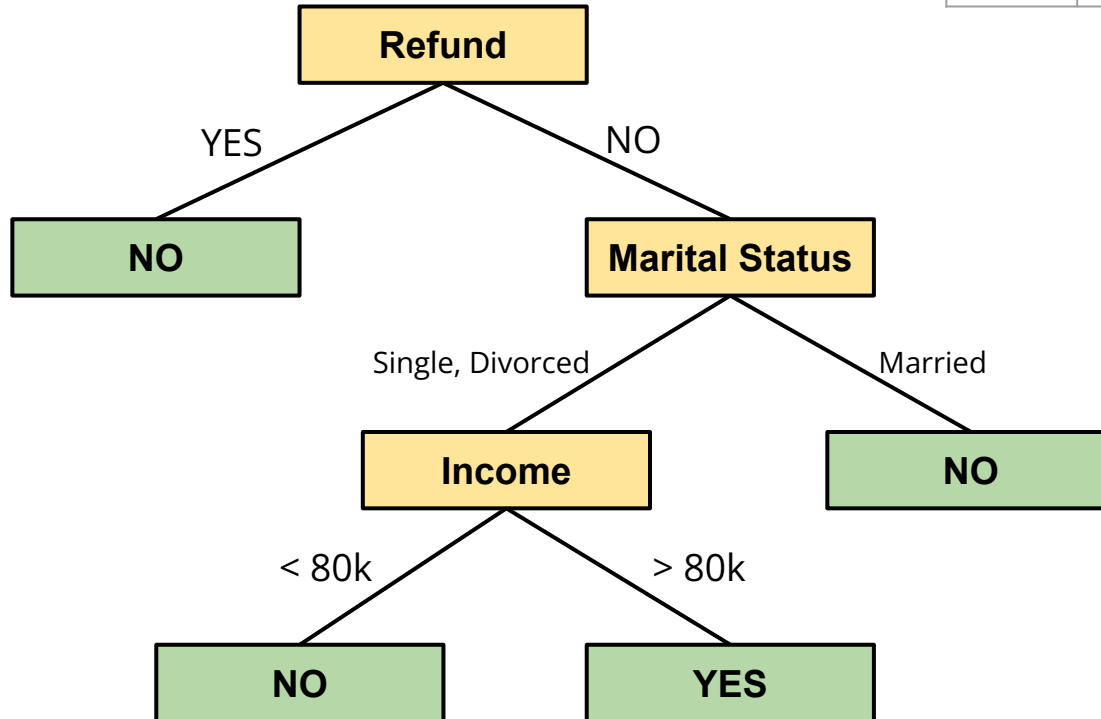
Refund	Marital Status	Income	Class
Yes	Single	125k	No
No	Married	100k	No
No	Single	70k	No
Yes	Married	120k	No
No	Divorced	90k	Yes
No	Married	60k	No
Yes	Divorced	220k	No
No	Single	85k	Yes
No	Married	75k	No
No	Single	90k	Yes

What a Decision Tree looks like

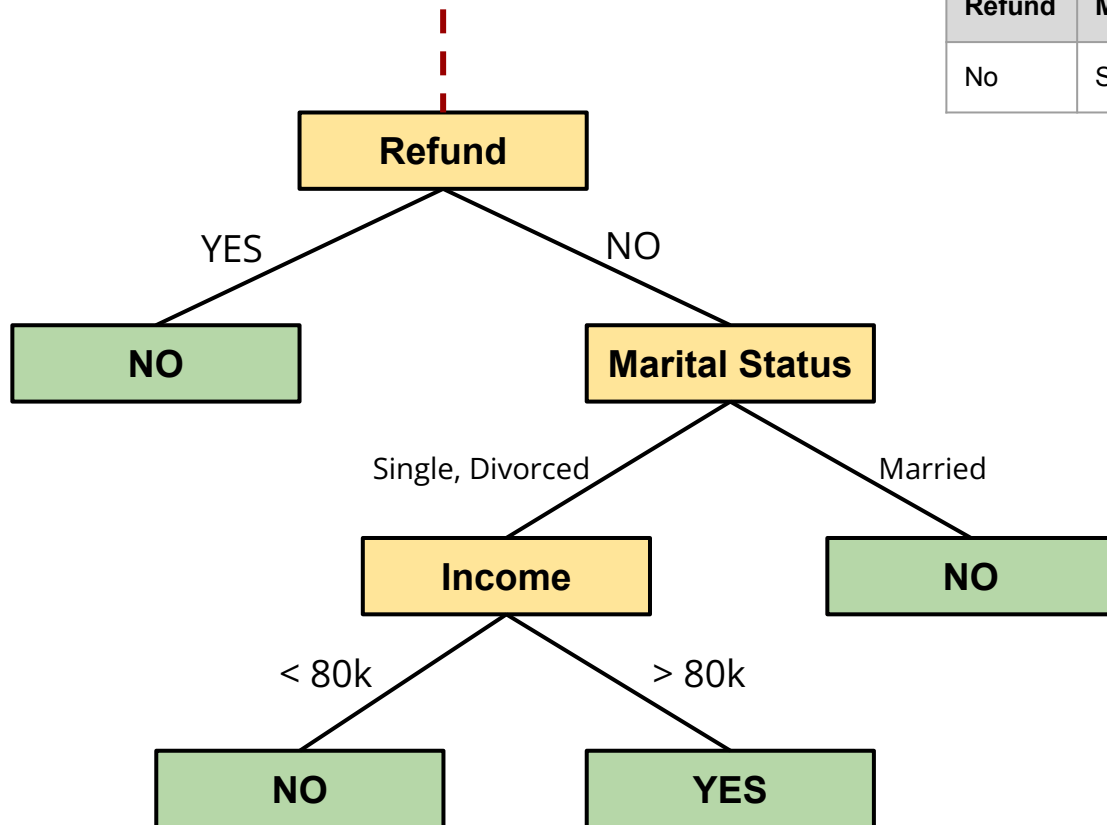


How it works

Refund	Marital Status	Income	Class
No	Single	70k	?

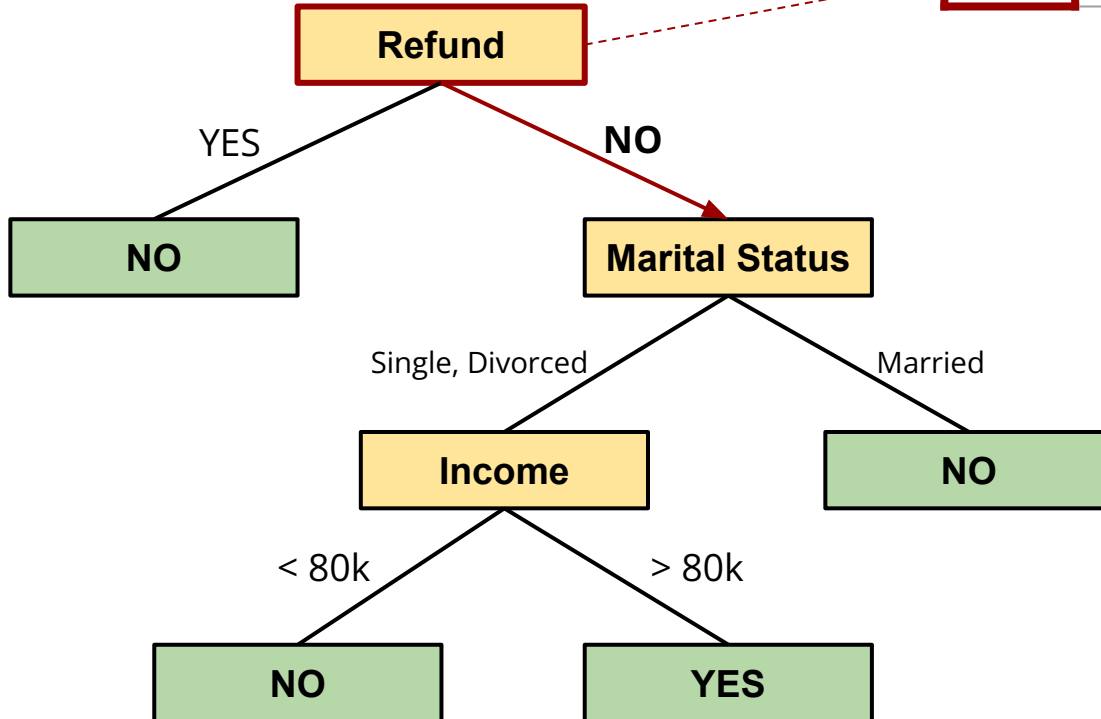


Start at root node

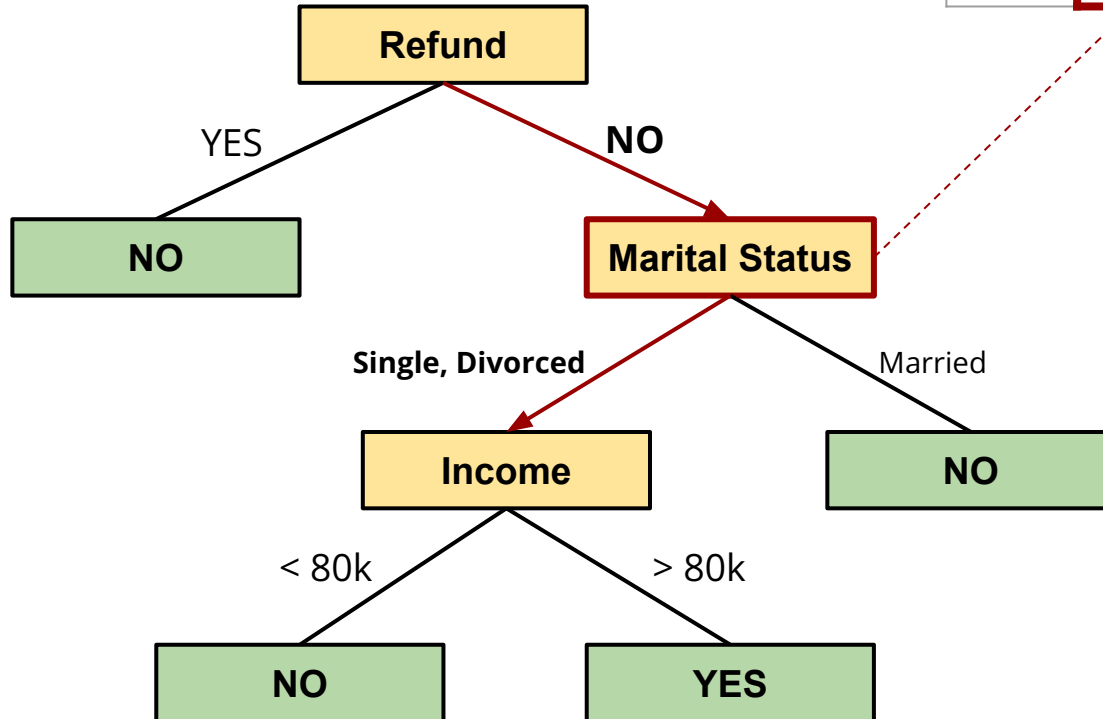


Refund	Marital Status	Income	Class
No	Single	70k	?

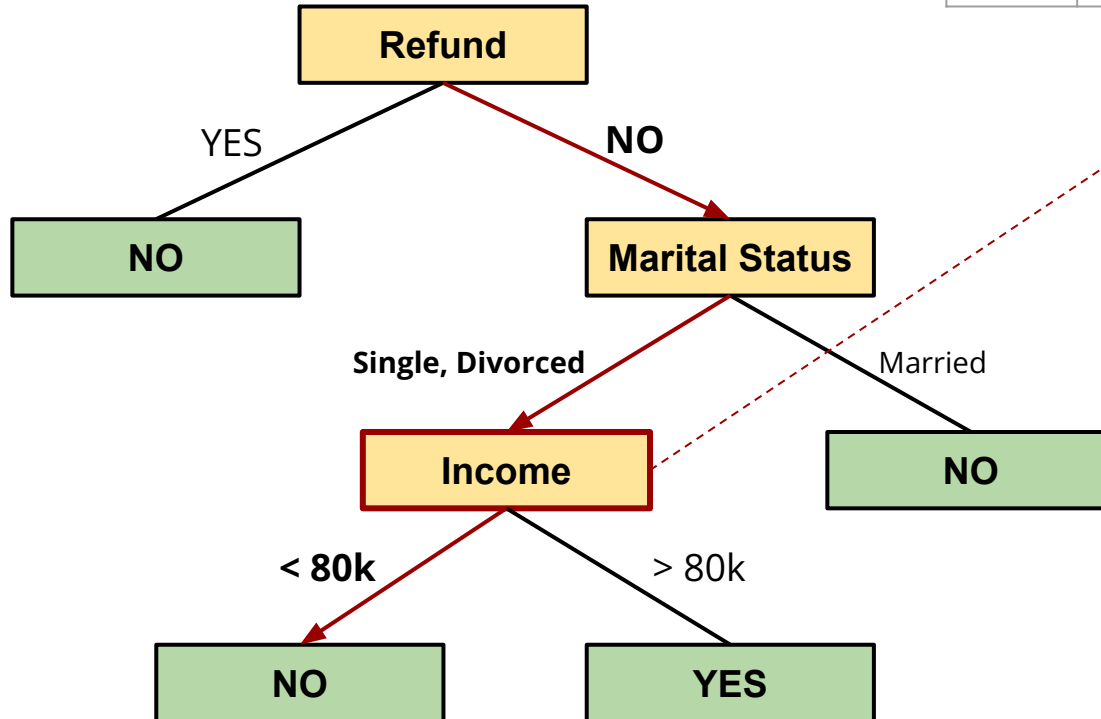
Refund	Marital Status	Income	Class
No	Single	70k	?



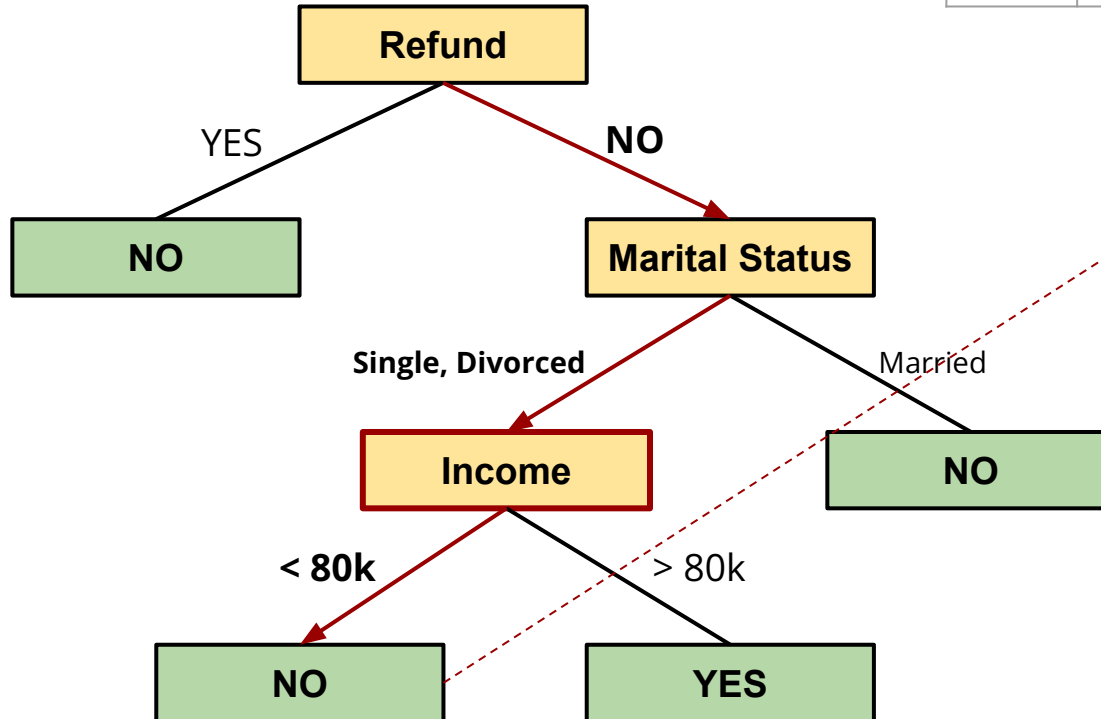
Refund	Marital Status	Income	Class
No	Single	70k	?



Refund	Marital Status	Income	Class
No	Single	70k	?



Refund	Marital Status	Income	Class
No	Single	70k	NO



Part 1

How do we learn it?

Refund	Marital Status	Income	Class
Yes	Single	125k	No
No	Married	100k	No
No	Single	70k	No
Yes	Married	120k	No
No	Divorced	90k	Yes
No	Married	60k	No
Yes	Divorced	220k	No
No	Single	85k	Yes
No	Married	75k	No
No	Single	90k	Yes

IF marital status == Married

Refund	Marital Status	Income	Class
Yes	Single	125k	No
No	Married	100k	No
No	Single	70k	No
Yes	Married	120k	No
No	Divorced	90k	Yes
No	Married	60k	No
Yes	Divorced	220k	No
No	Single	85k	Yes
No	Married	75k	No
No	Single	90k	Yes

IF marital status == Married

Refund	Marital Status	Income	Class
No	Married	100k	No
Yes	Married	120k	No
No	Married	60k	No
No	Married	75k	No

THEN class = NO

Refund	Marital Status	Income	Class
Yes	Single	125k	No
No	Married	100k	No
No	Single	70k	No
Yes	Married	120k	No
No	Divorced	90k	Yes
No	Married	60k	No
Yes	Divorced	220k	No
No	Single	85k	Yes
No	Married	75k	No
No	Single	90k	Yes

IF income < 60k

Refund	Marital Status	Income	Class
Yes	Single	125k	No
No	Married	100k	No
No	Single	70k	No
Yes	Married	120k	No
No	Divorced	90k	Yes
No	Married	60k	No
Yes	Divorced	220k	No
No	Single	85k	Yes
No	Married	75k	No
No	Single	90k	Yes

IF income < 60k

Refund	Marital Status	Income	Class
--------	----------------	--------	-------

THEN ?

we've never seen an example where this condition is true,
then we can predict the majority class, etc. --> "predict a reasonable default"

Hunt's Algorithm

- Recursive Algorithm
 - Repeatedly split the dataset based on attributes
- Base cases:
 - IF Split and all data points in the same class
 - Great! Predict that class
 - IF Split and no data points
 - No problem! Predict a reasonable default

Hunt's Algorithm

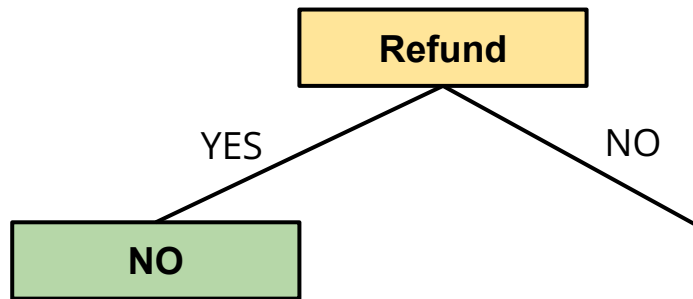
The recursion (IF split and data points belong to more than one class)

- Find the attribute (and best way to split that attribute) that best splits the data

Example

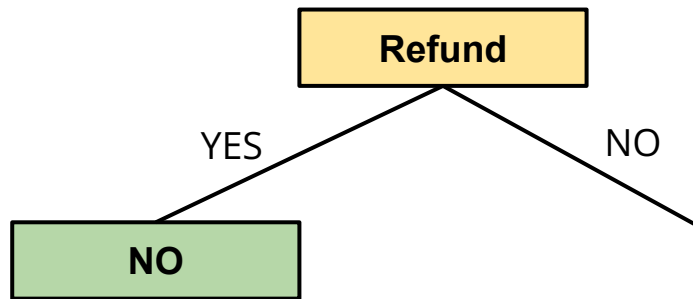
Refund	Marital Status	Income	Class
Yes	Single	125k	No
No	Married	100k	No
No	Single	70k	No
Yes	Married	120k	No
No	Divorced	90k	Yes
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Yes	Divorced	220k	No
No	Single	85k	Yes
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Yes	Divorced	220k	No
No	Single	85k	Yes
No	Married	75k	No
No	Single	90k	Yes



Refund	Marital Status	Income	Class
Yes	Single	125k	No
Yes	Married	120k	No
Yes	Divorced	220k	No

Refund	Marital Status	Income	Class
No	Married	100k	No
No	Single	70k	No
No	Divorced	90k	Yes
No	Married	60k	No
No	Single	85k	Yes
No	Married	75k	No
No	Single	90k	Yes

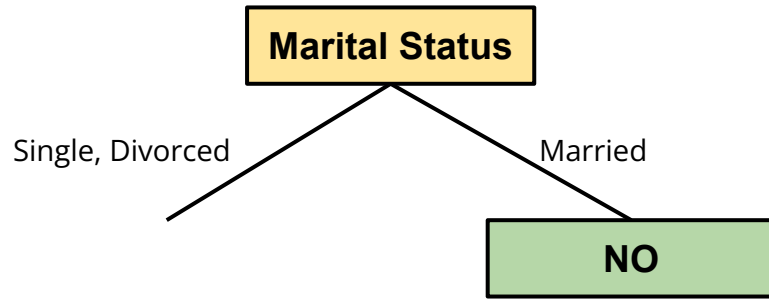


Refund	Marital Status	Income	Class
Yes	Single	125k	No
Yes	Married	120k	No
Yes	Divorced	220k	No

Refund	Marital Status	Income	Class
No	Married	100k	No
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No	Married	60k	No
No	Single	85k	Yes
No	Married	75k	No
No	Single	90k	Yes

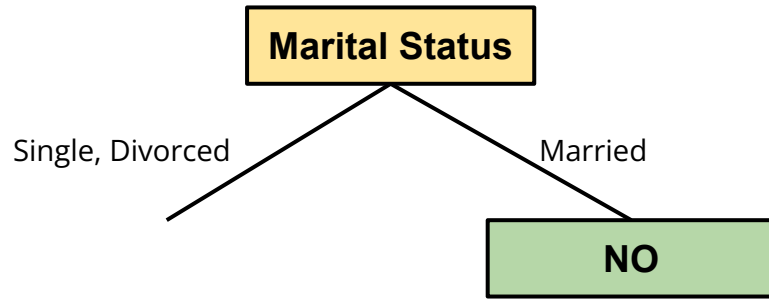
Refund	Marital Status	Income	Class
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No	Single	70k	No
No	Divorced	90k	Yes
No	Married	60k	No
No	Single	85k	Yes
No	Married	75k	No
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No	Single	70k	No
No	Divorced	90k	Yes
No	Married	60k	No
No	Single	85k	Yes
No	Married	75k	No
No	Single	90k	Yes



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No	Single	70k	No
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No	Single	85k	Yes
No	Single	90k	Yes

Refund	Marital Status	Income	Class
No	Married	100k	No
No	Married	60k	No
No	Married	75k	No

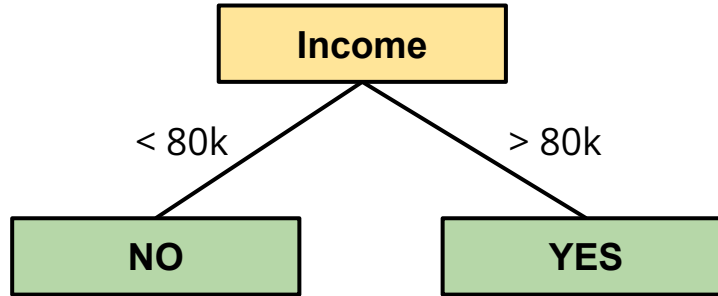


Refund	Marital Status	Income	Class
No	Single	70k	No
No	Divorced	90k	Yes
No	Single	85k	Yes
No	Single	90k	Yes

Refund	Marital Status	Income	Class
No	Married	100k	No
No	Married	60k	No
No	Married	75k	No

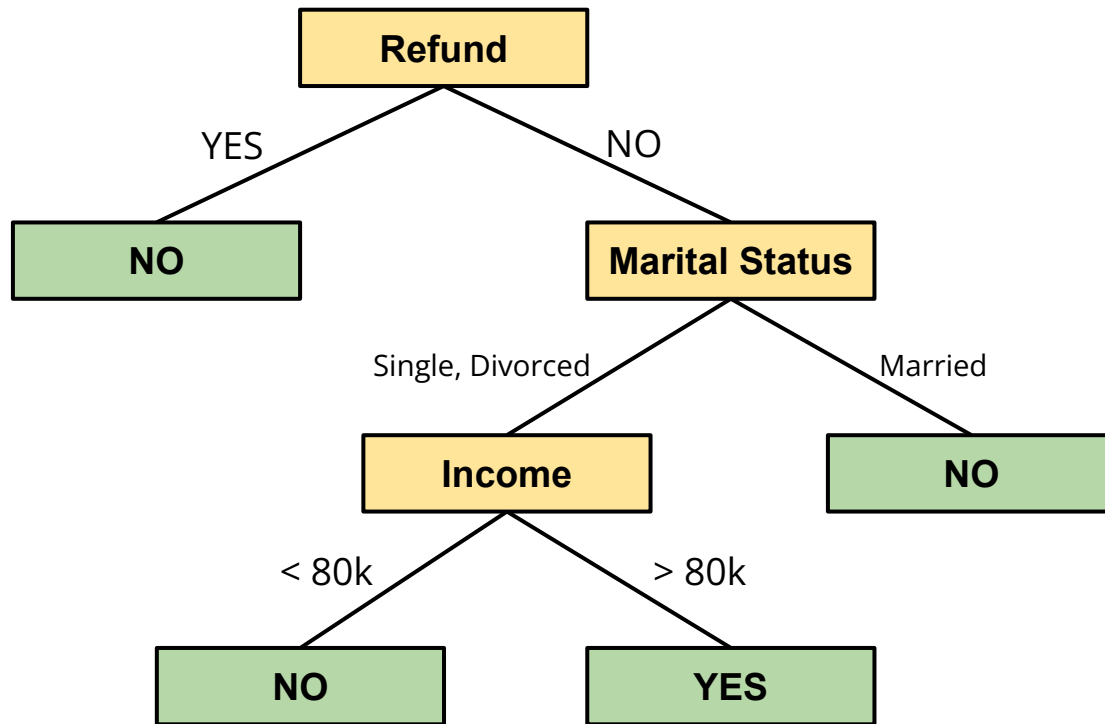
Refund	Marital Status	Income	Class
No	Single	70k	No
No	Divorced	90k	Yes
No	Single	85k	Yes
No	Single	90k	Yes

Refund	Marital Status	Income	Class
No	Single	70k	No
No	Divorced	90k	Yes
No	Single	85k	Yes
No	Single	90k	Yes



Refund	Marital Status	Income	Class
No	Single	70k	No

Refund	Marital Status	Income	Class
No	Divorced	90k	Yes
No	Single	85k	Yes
No	Single	90k	Yes

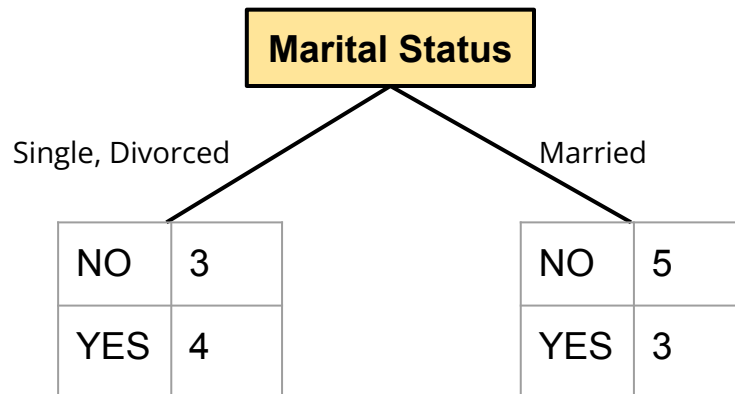
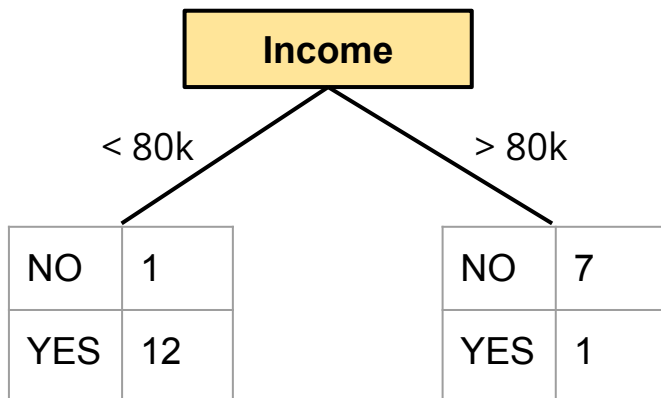


What do we mean by best split?

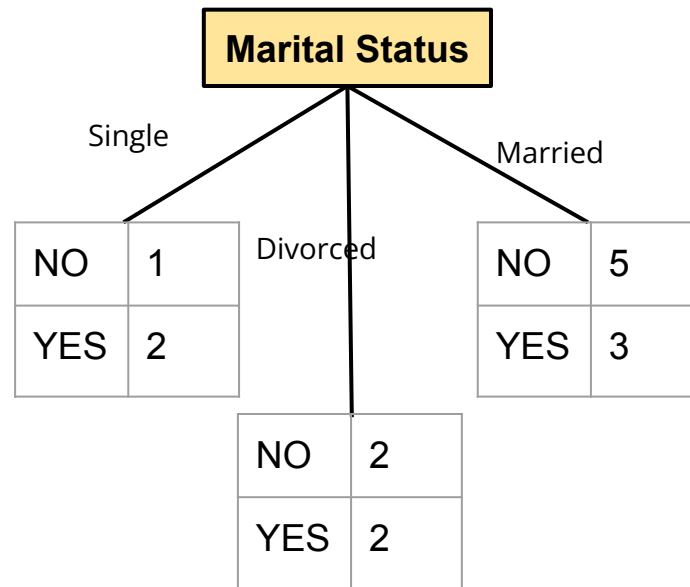
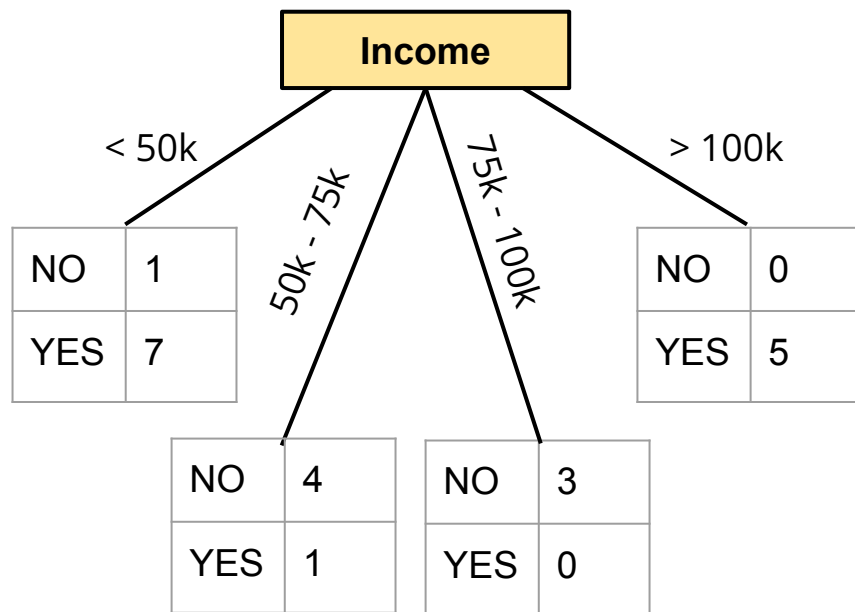
Many ways to split a given attribute

- Binary Split
- Multi-Way Split

Binary Split



Multi-Way Split



Continuous Variables

- Use binning before running the decision tree
 - Can use clustering for that for example
- Compute a threshold while building the tree
 - $A > t$ vs $A < t$

Need a metric

That favors nodes like this:

NO	1
YES	7

Over nodes like this:

NO	4
YES	4

GINI index

Denote $p(j \mid t)$ as the relative frequency of class j at node t .

NO	1
YES	7

$$p(\text{NO} \mid t) = \frac{1}{8}$$

$$p(\text{YES} \mid t) = \frac{7}{8}$$

NO	4
YES	3

$$p(\text{NO} \mid t) = \frac{4}{7}$$

$$p(\text{YES} \mid t) = \frac{3}{7}$$

GINI index

$$GINI(t) = 1 - \sum_j p(j|t)^2$$

NO	1
YES	7

$$p(\text{NO} \mid t) = 1/8$$

$$p(\text{YES} \mid t) = 7/8$$

$$GINI(t) = 1 - 1/64 - 49/64 = 14/64$$

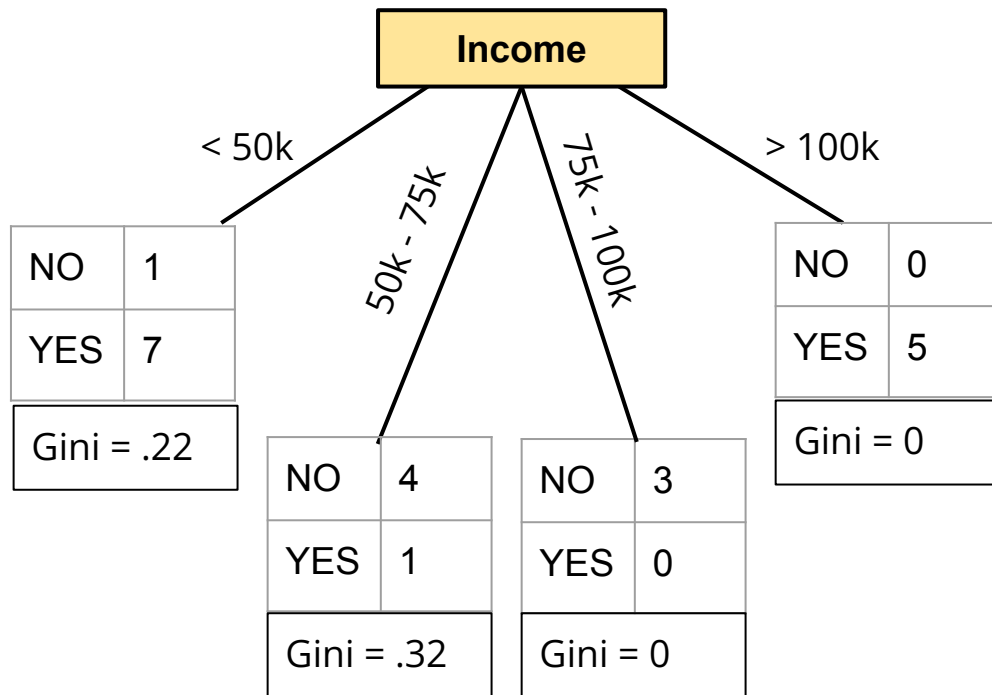
NO	4
YES	3

$$p(\text{NO} \mid t) = 4/7$$

$$p(\text{YES} \mid t) = 3/7$$

$$GINI(t) = 1 - 16/49 - 9/49 = 24/49$$

GINI of the Split



GINI of the split

$$GINI_{split} = \sum_{t=1}^k \frac{n_t}{n} GINI(t)$$

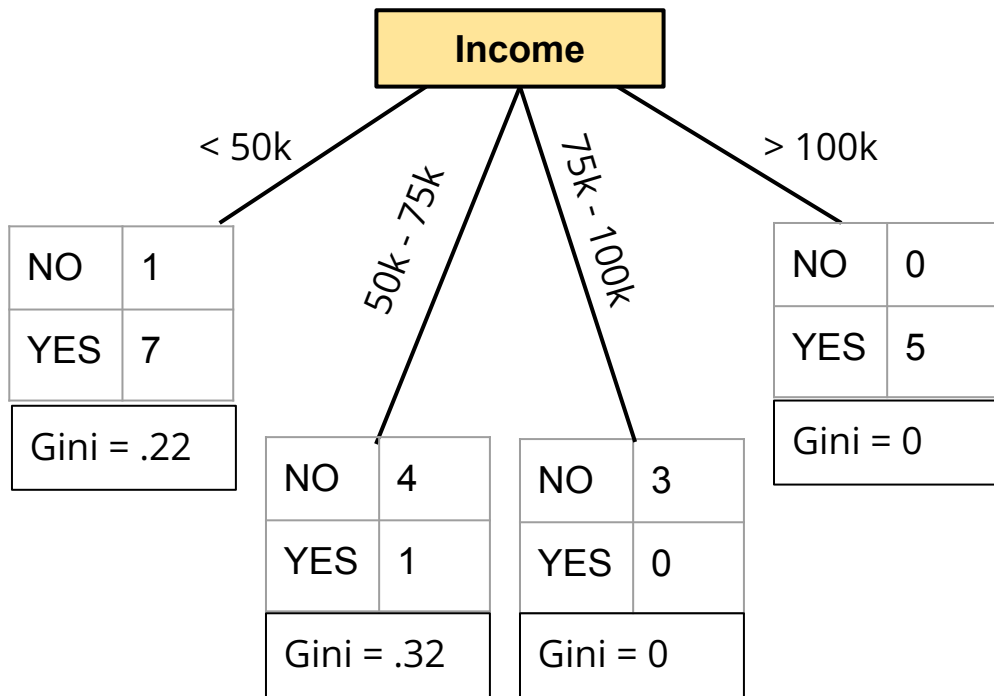
where:

n_t = number of data points at node t

n = number of data points before the split (parent node)

GINI of the split

weighted average



$$GINI_{split} = \sum_{t=1}^k \frac{n_t}{n} GINI(t)$$

$$n = 21$$

$$\begin{aligned} GINI_{split} &= .22 * 8/21 \\ &\quad + .32 * 5/21 \\ &\quad + 0 * 3/21 \\ &\quad + 0 * 5/21 \\ &= .16 \end{aligned}$$

Part 2

Putting it all together

Before splitting

NO	8
YES	7
Gini = .49	

Before splitting

NO	8
YES	7
Gini = .49	

.50 is the max
val of GINI

Income

< 80k

> 80k

NO	1
YES	6

NO	7
YES	1

Marital Status

Single

Divorced

Married

NO	1
YES	2

NO	2
YES	2

NO	5
YES	3

Before splitting

NO	8
YES	7
Gini = .49	

Income

< 80k

> 80k

NO	1
YES	6
Gini = .24	

NO	7
YES	1
Gini = .22	

Marital Status

Single

Divorced

Married

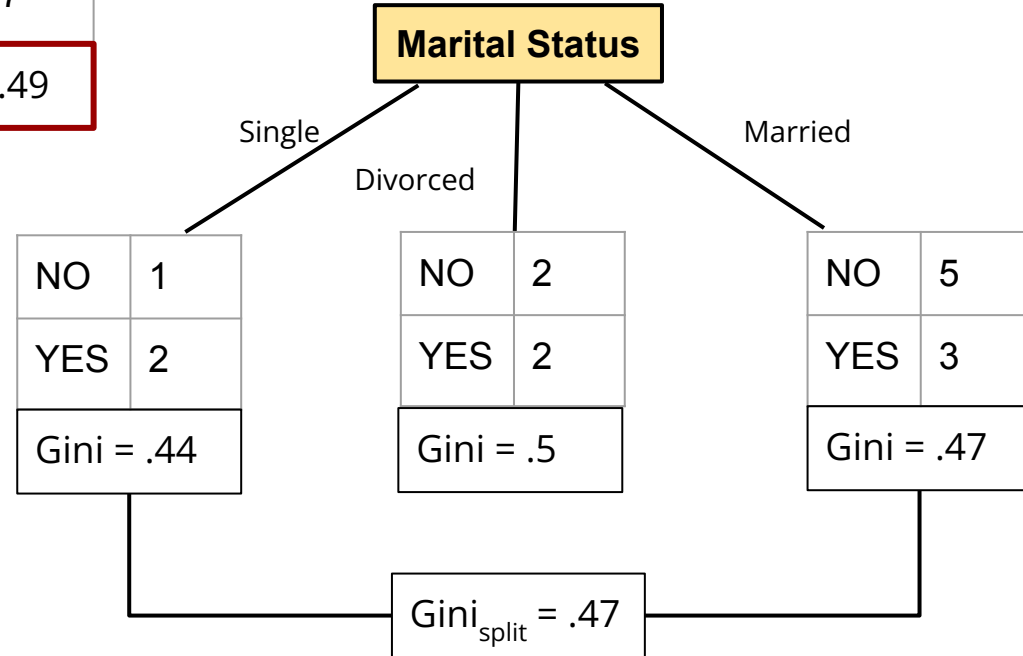
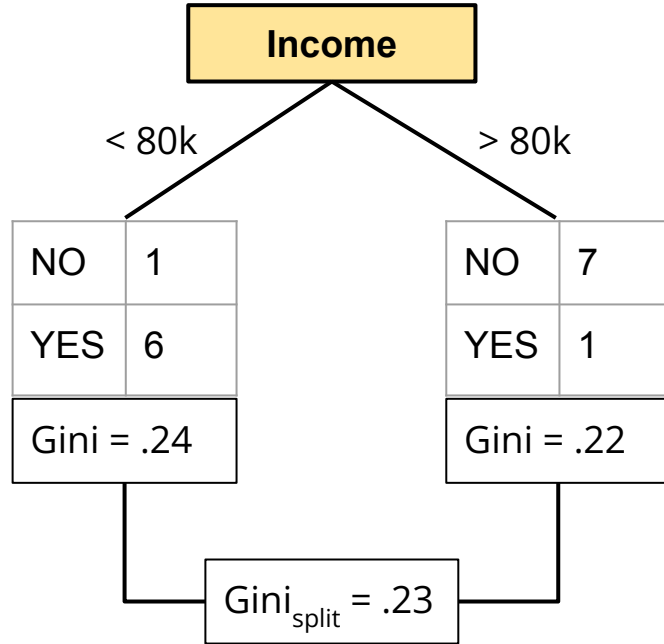
NO	1
YES	2
Gini = .44	

NO	2
YES	2
Gini = .5	

NO	5
YES	3
Gini = .47	

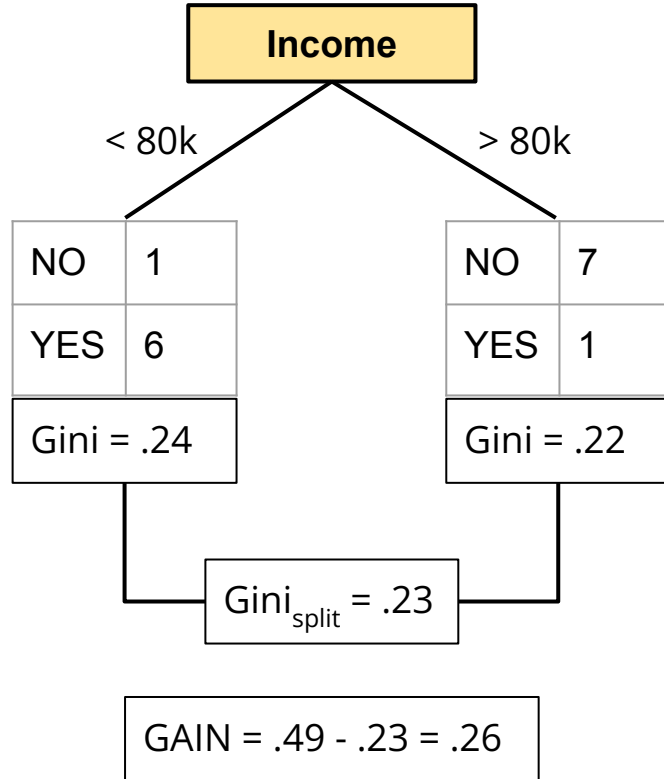
Before splitting

NO	8
YES	7
Gini = .49	

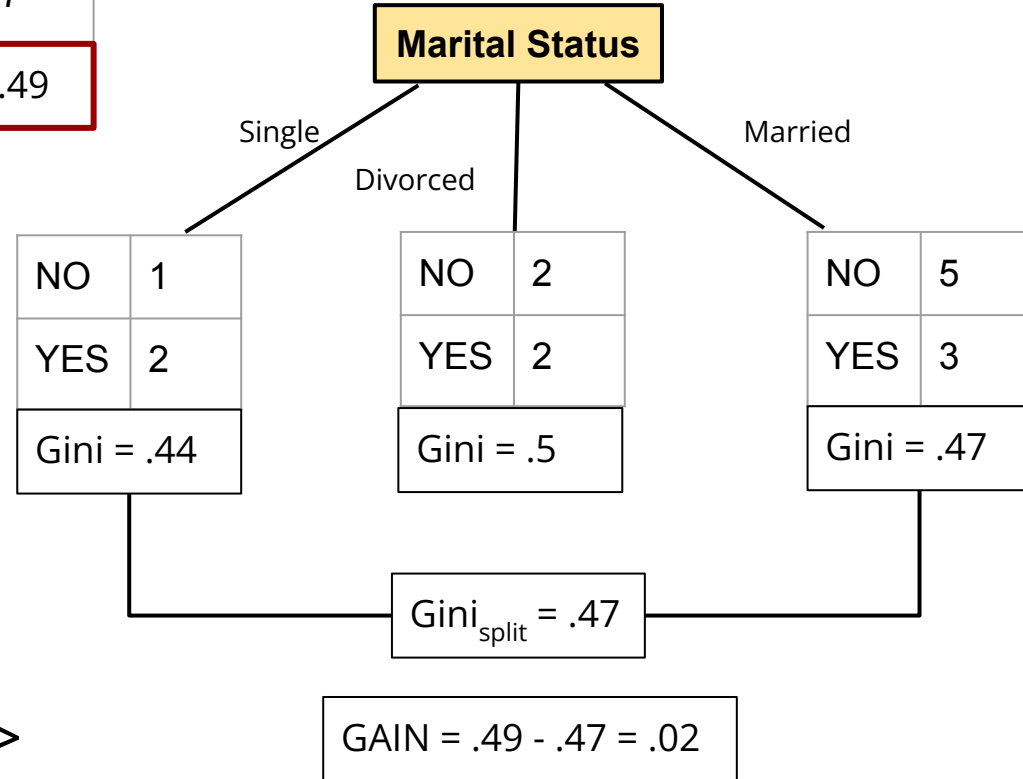


Before splitting

NO	8
YES	7
Gini = .49	

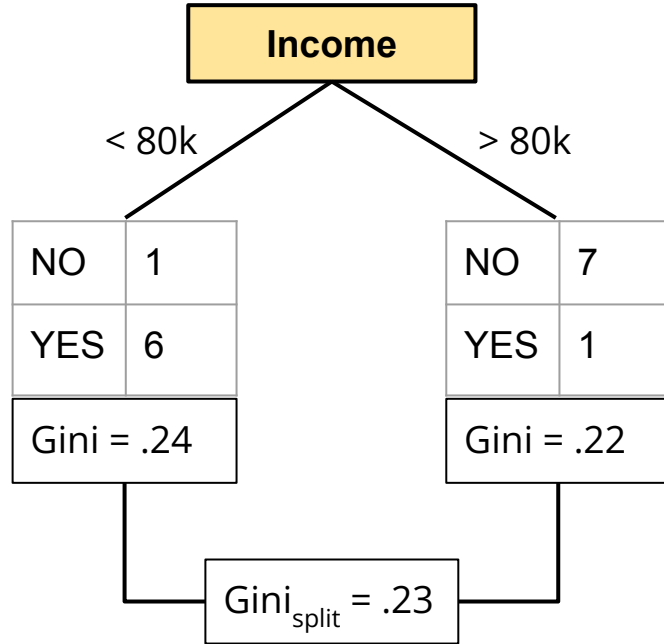


>



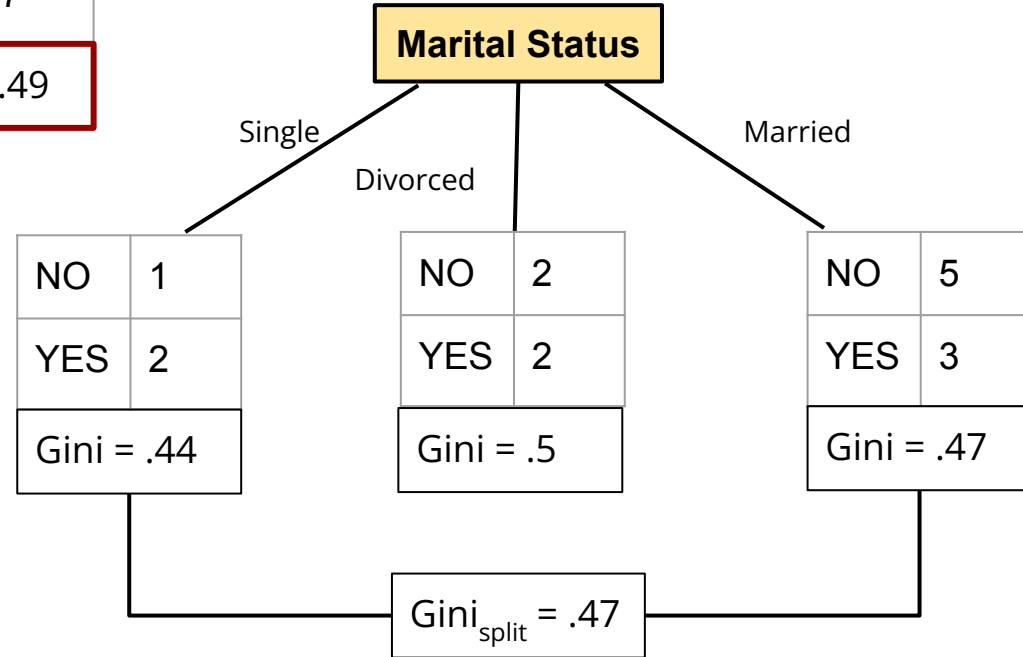
Before splitting

NO	8
YES	7
Gini = .49	



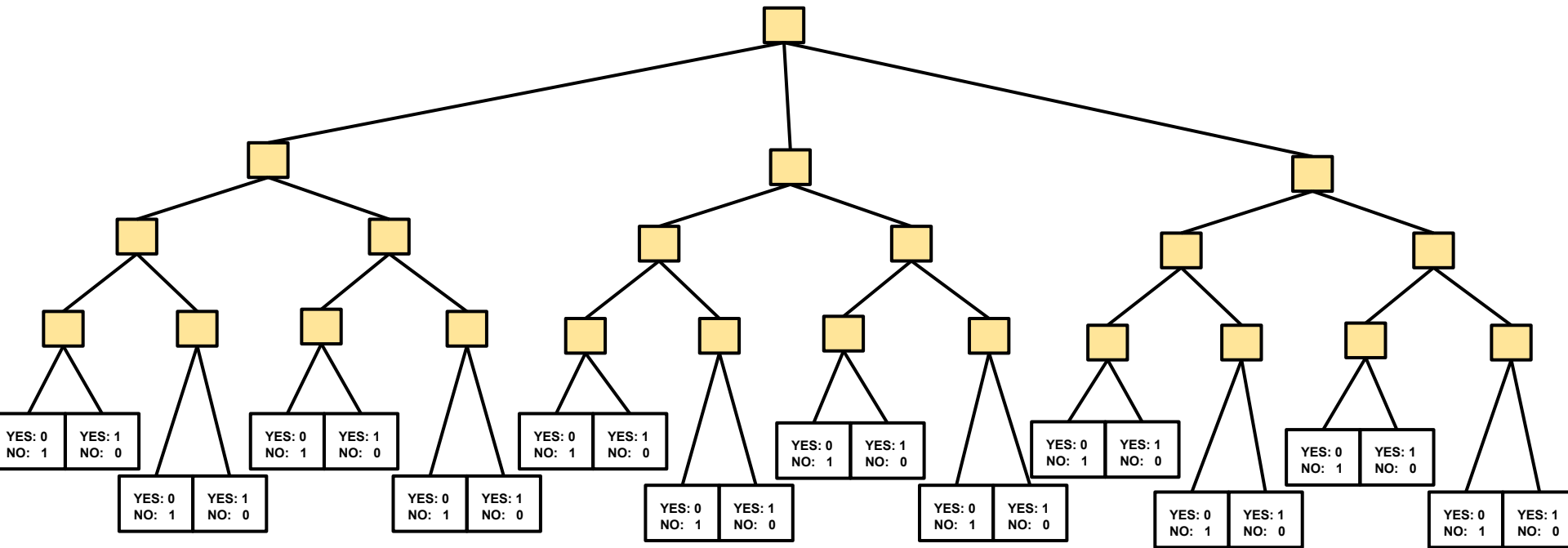
$$\text{GAIN} = .49 - .23 = .26$$

>



$$\text{GAIN} = .49 - .47 = .02$$

Limitations



Easy to construct a tree that is too complex and overfits the data.

Solutions:

- Early termination (stop before tree is fully grown - use majority vote at leaf node)
 - Stop at some specified depth
 - Stop if size of node is below some threshold
 - Stop if gini does not improve
- Pruning (create fully grown tree then trim)

Extensions

Other measures of node purity

- Entropy

$$\text{Entropy}(t) = - \sum_j p(j|t) \log(p(j|t))$$

- Misclassification Error

$$\text{Error}(t) = 1 - \max_j (p(j|t))$$

Part 3